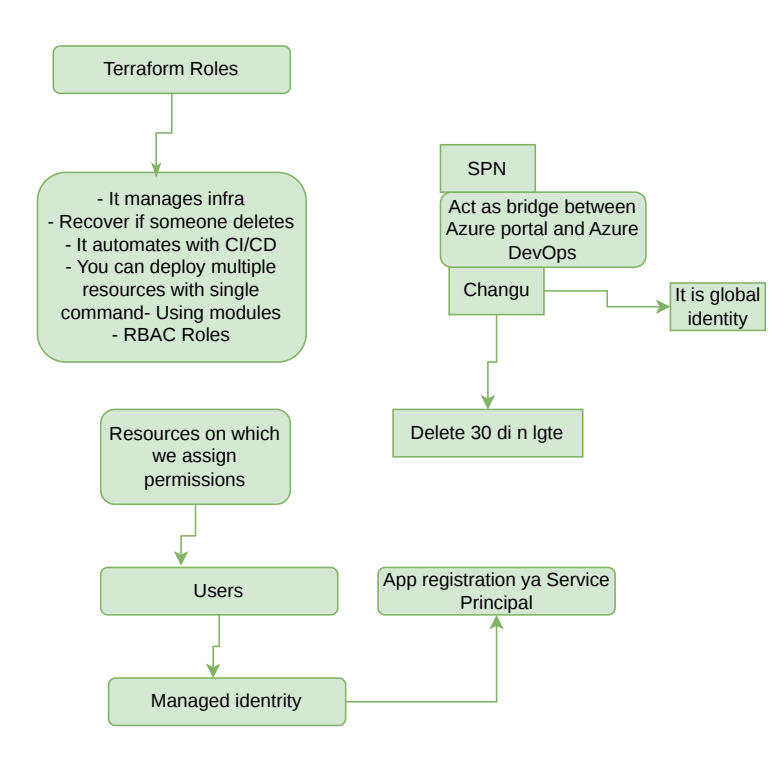
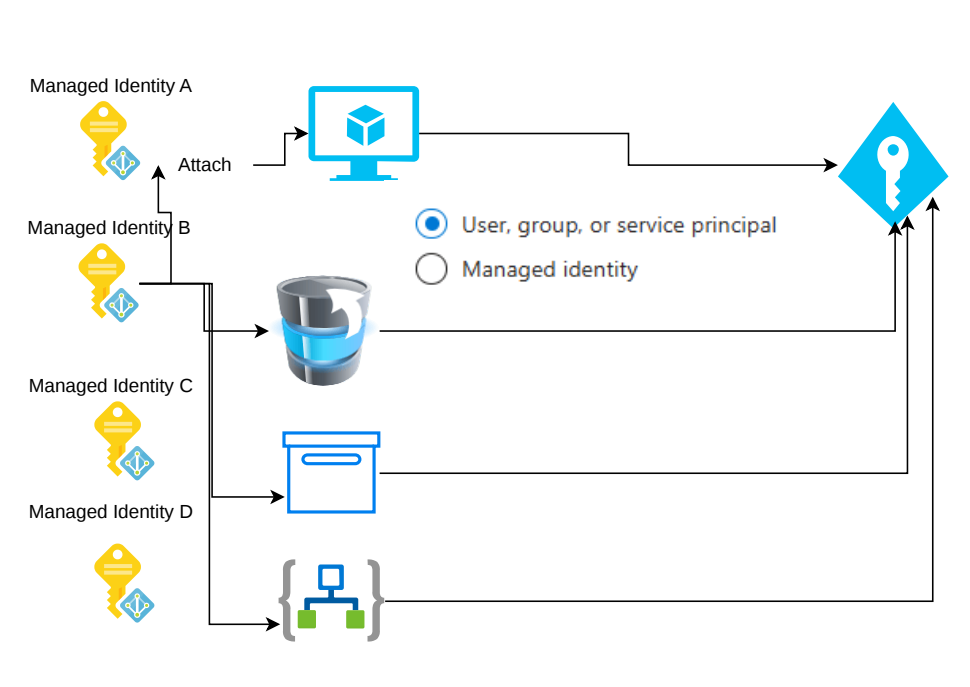
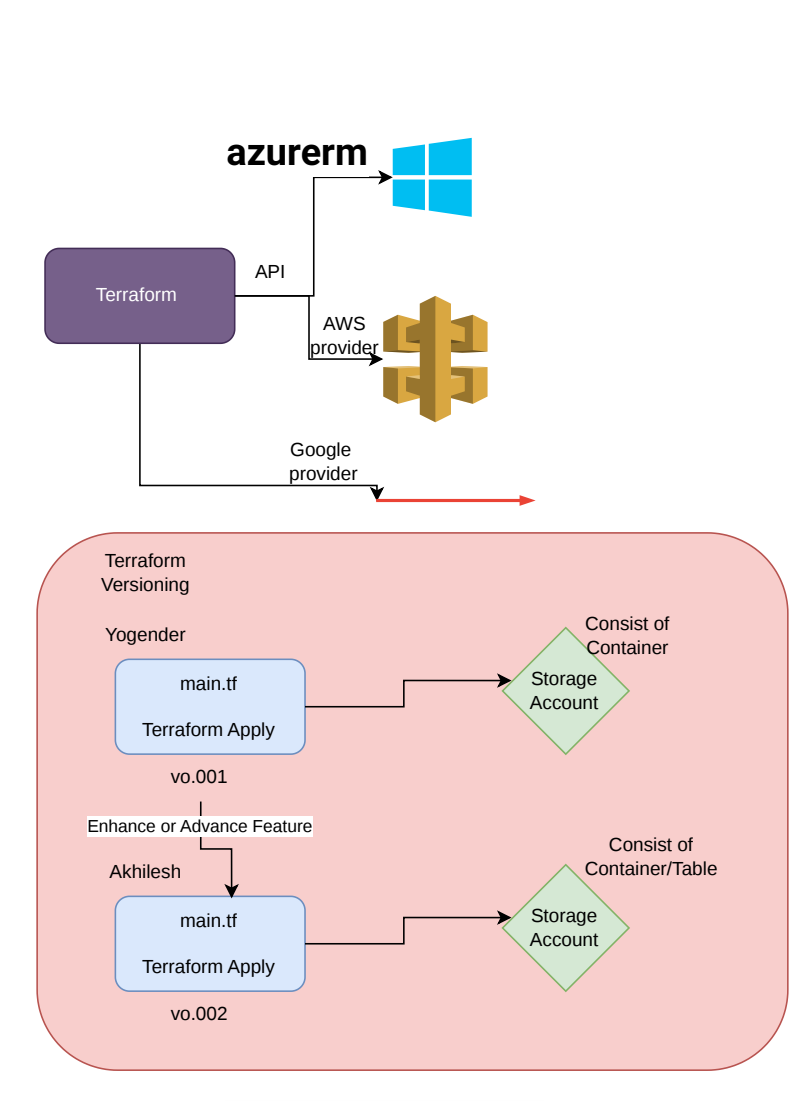

Terraform Roles
- It manages infra
- Recover if someone deletes
- It automates with CI/CD
- You can deploy multiple resources with single command- Using modules
- RBAC Roles
SPN
Act as bridge between Azure portal and Azure DevOps
Changu
It is global identity
Delete 30 di n lgte
Resources on which we assign permissions
Users
App registration ya Service Principal
Managed identrity


Managed Identity A
Attach
Managed Identity B
● User, group, or service principal
○ Managed identity
Managed Identity C
Managed Identity D

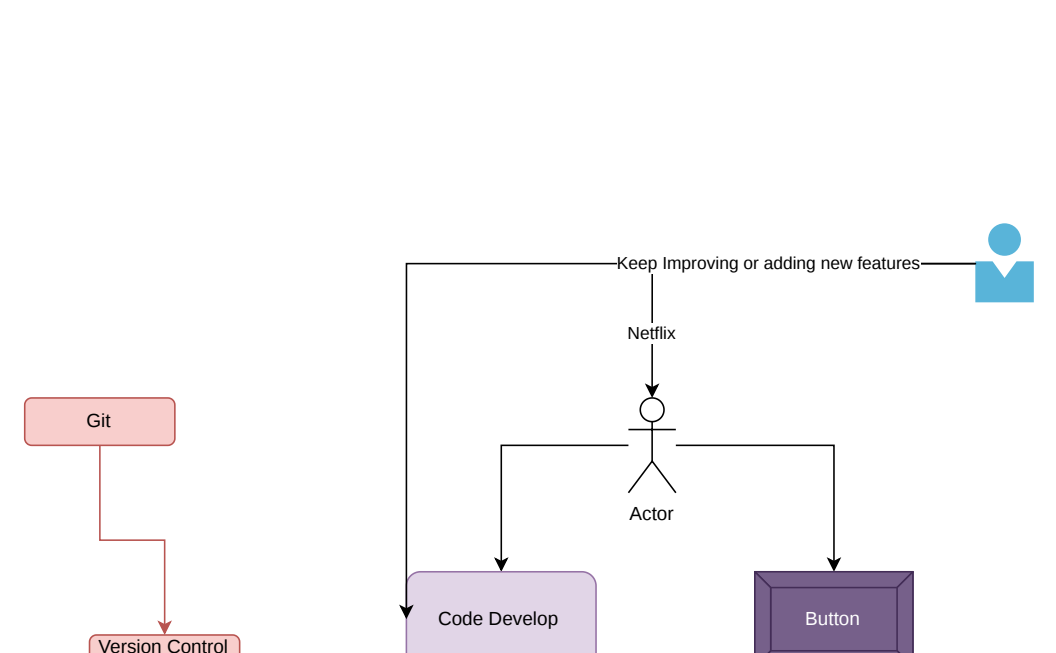

azurerm
Terraform
API
AWS provider
Google provider
Terraform Versioning
Yogender
main.tf
Terraform Apply
vo.001
Consist of Container
Storage Account
Enhance or Advance Feature
Akhilesh
main.tf
Terraform Apply
vo.002
Consist of Container/Table
Storage Account

Code is Everything

tfstate - brain of Terraform resource - it has all detailes about creation and deletion

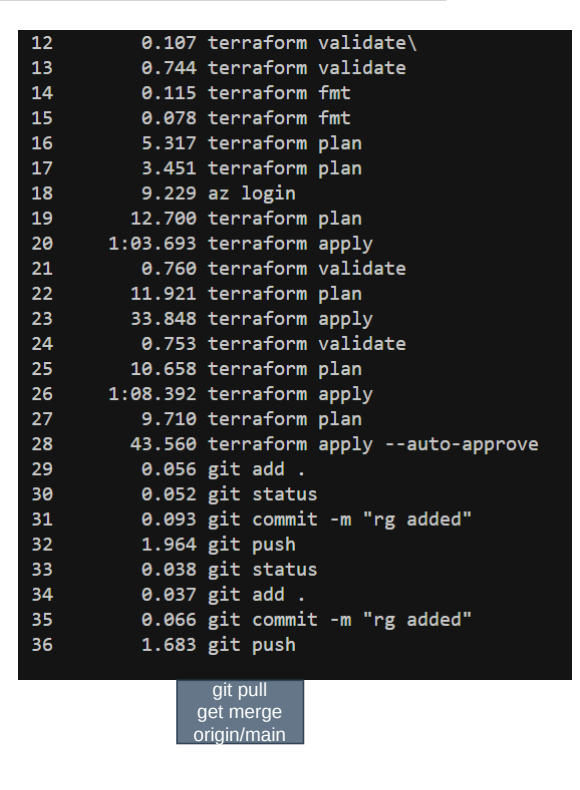
Arguments- Properties which allows to handle the configuration of that resource

Always create the resource manually first from Portal


Keep Improving or adding new features
Netflix
Git
Version Control System
Actor
Code Develop
Button
V1.01
V1.02
Git tech or platform

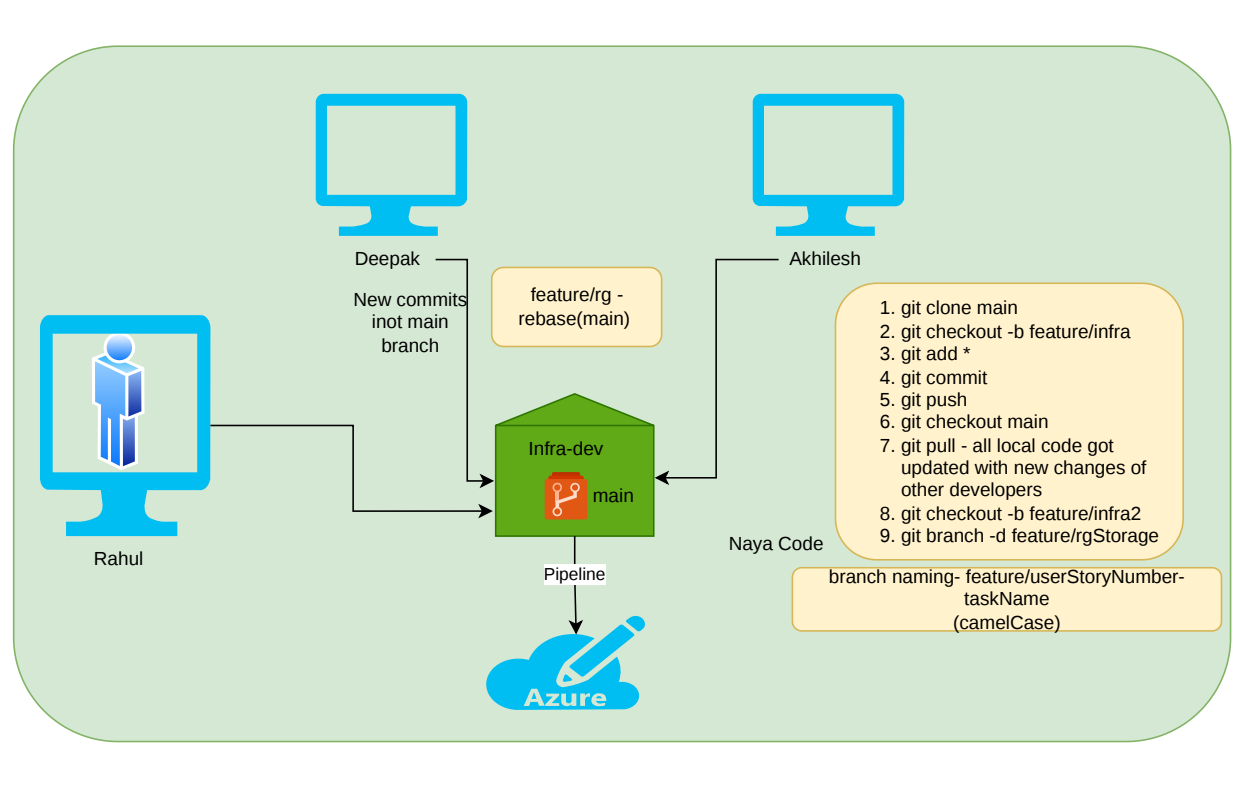
git clone ----address

terraform fmt = shift+alt+f



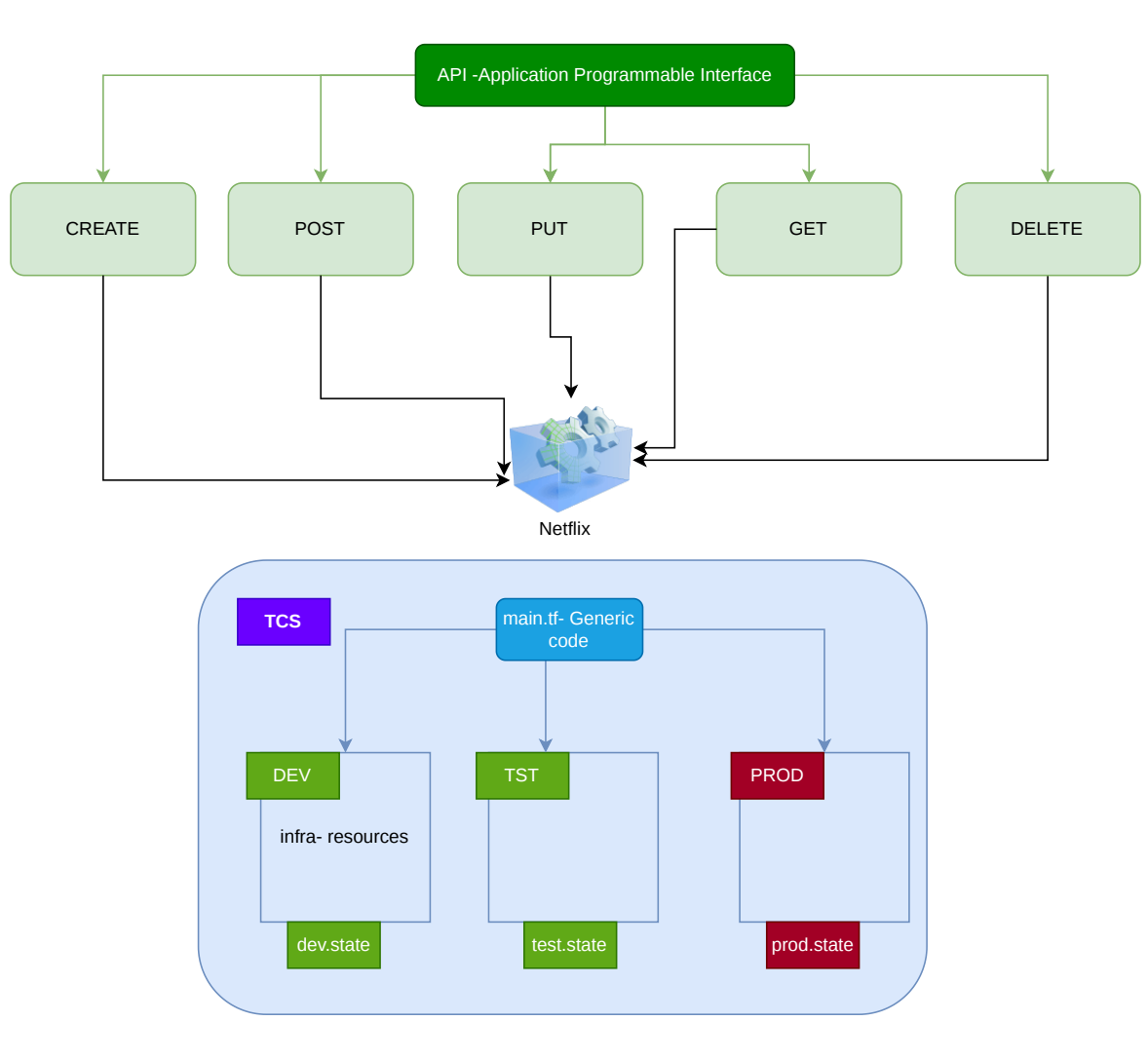
```
12       0.107 terraform validate\
13       0.744 terraform validate
14       0.115 terraform fmt
15       0.078 terraform fmt
16       5.317 terraform plan
17       3.451 terraform plan
18       9.229 az login
19      12.700 terraform plan
20    1:03.693 terraform apply
21       0.760 terraform validate
22      11.921 terraform plan
23      33.848 terraform apply
24       0.753 terraform validate
25      10.658 terraform plan
26    1:08.392 terraform apply
27       9.710 terraform plan
28      43.560 terraform apply --auto-approve
29       0.056 git add .
30       0.052 git status
31       0.093 git commit -m "rg added"
32       1.964 git push
33       0.038 git status
34       0.037 git add .
35       0.066 git commit -m "rg added"
36       1.683 git push
```

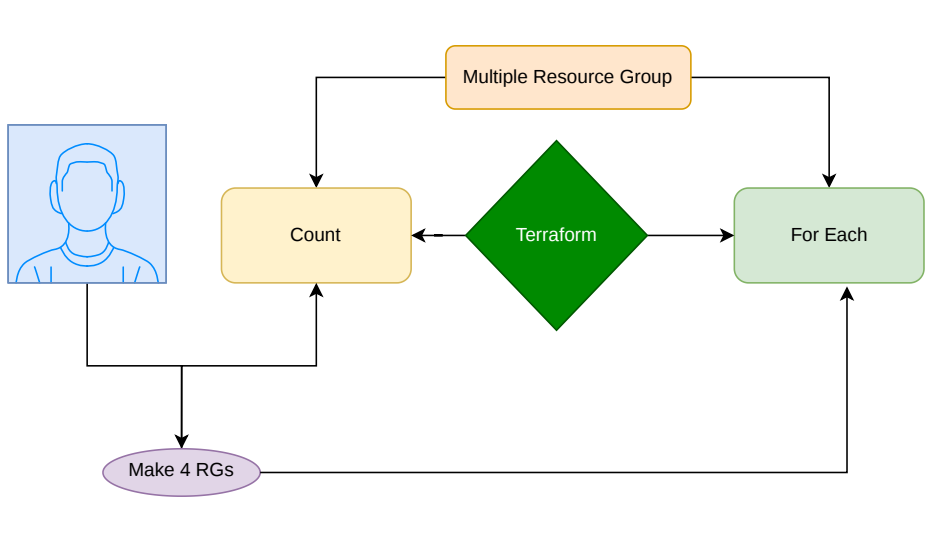
git pull
get merge origin/main


Deepak
New commits inot main branch
feature/rg - rebase(main)
Akhilesh
Rahul
Infra-dev
main
Pipeline
Azure
Naya Code

1. git clone main
2. git checkout -b feature/infra
3. git add *
4. git commit
5. git push
6. git checkout main
7. git pull - all local code got updated with new changes of other developers
8. git checkout -b feature/infra2
9. git branch -d feature/rgStorage

branch naming- feature/userStoryNumber-taskName (camelCase)


API -Application Programmable Interface
CREATE
POST
PUT
GET
DELETE
Netflix
TCS
main.tf- Generic code
DEV
infra- resources
TST
PROD
dev.state
test.state
prod.state


Multiple Resource Group
Count
Terraform
For Each
Make 4 RGs

Chaining

Count

Count allows to deploy multiple resources using counts i.e.
Server1, Server2, Server3.....
User1, User2, User3.......

"Vegetable basket"= {
"Red Vegetable" = {"capsicum" = "250gm", "chillies" = "100gm"},
"Green Vegetables" = {"spinach" = "2kg", "cucumber"= 3, "bottle guard" = "4kg", "lady finger"= "1kg"}
}

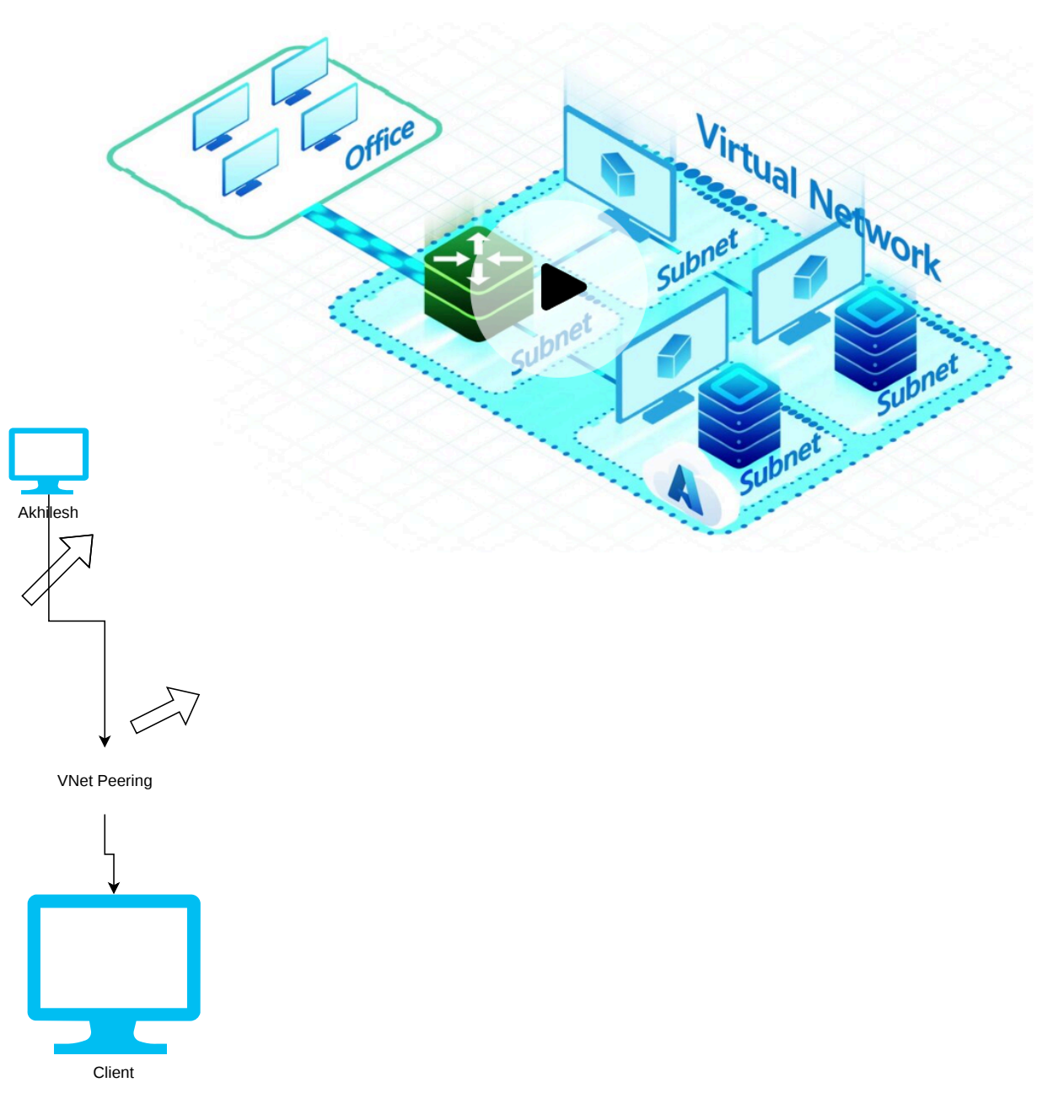
- vegetable basket.redVegetabel = capsicum, chillies
- vegetableBasket.greenVegetables = spinach, cucumber, bottle guard, lady finger
- vegetable basket.redVegetabel.capsicum = 250gm

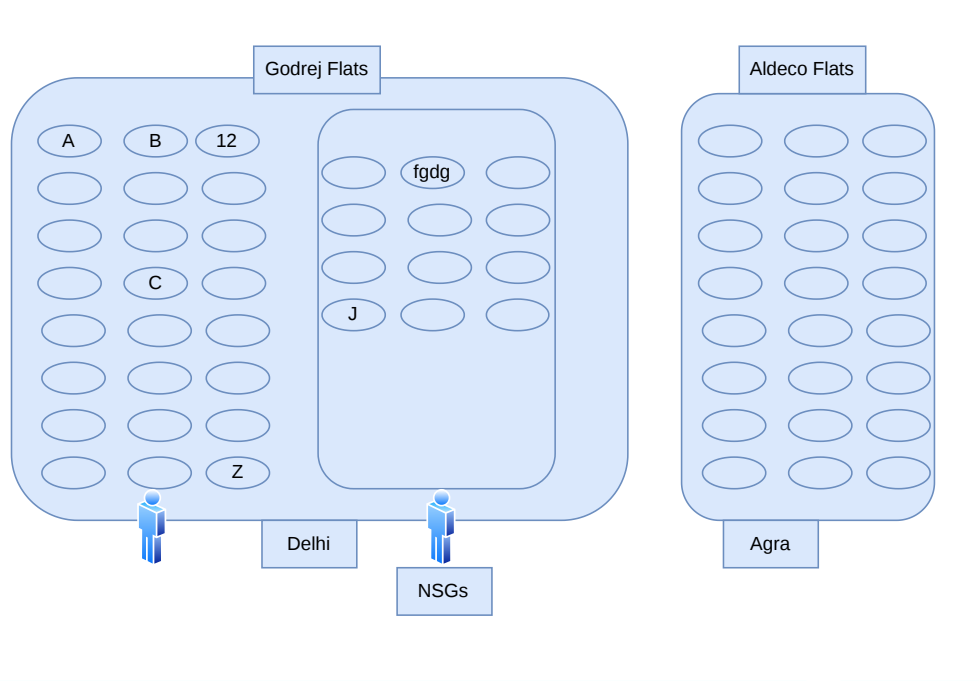
For Each

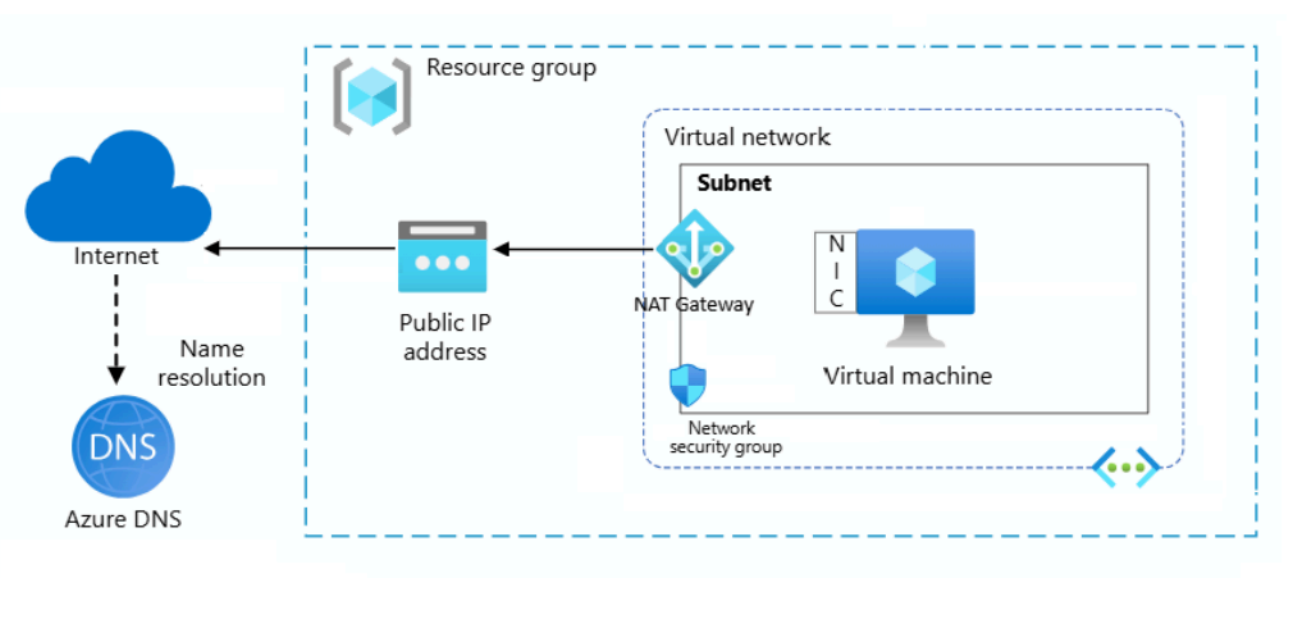
Loop that reads variables defined values as 'Key' or 'Values' form.

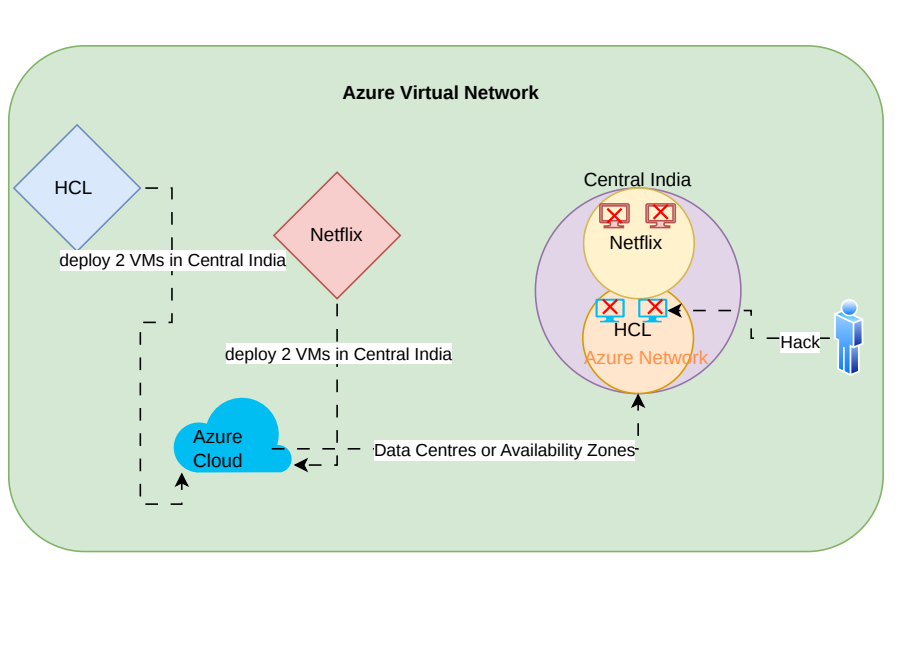
Key = Value

LHS RHS

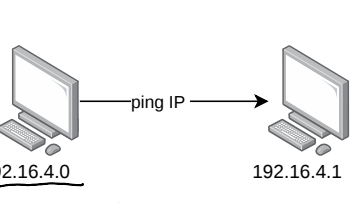

Office
Virtual Network
Subnet
Subnet
Subnet
Subnet
Akhilesh
VNet Peering
Client


Godrej Flats
A
B
12
C
Z
fgdg
J
Delhi
NSGs
Aldeco Flats
Agra


Resource group
Virtual network
Subnet
Internet
Public IP address
NAT Gateway
N I C
Virtual machine
Name resolution
DNS
Azure DNS
Network security group


Azure Virtual Network
HCL
deploy 2 VMs in Central India
Netflix
deploy 2 VMs in Central India
Central India
Netflix
HCL
Azure Network
"Hack"
Azure Cloud
Data Centres or Availability Zones

IP Addresses


ping IP
192.16.4.0
192.16.4.1

IPV4

IPV6

IPV4

192.16.4.1

4 bytes

1 byte = 8 bits

Total bits = 32

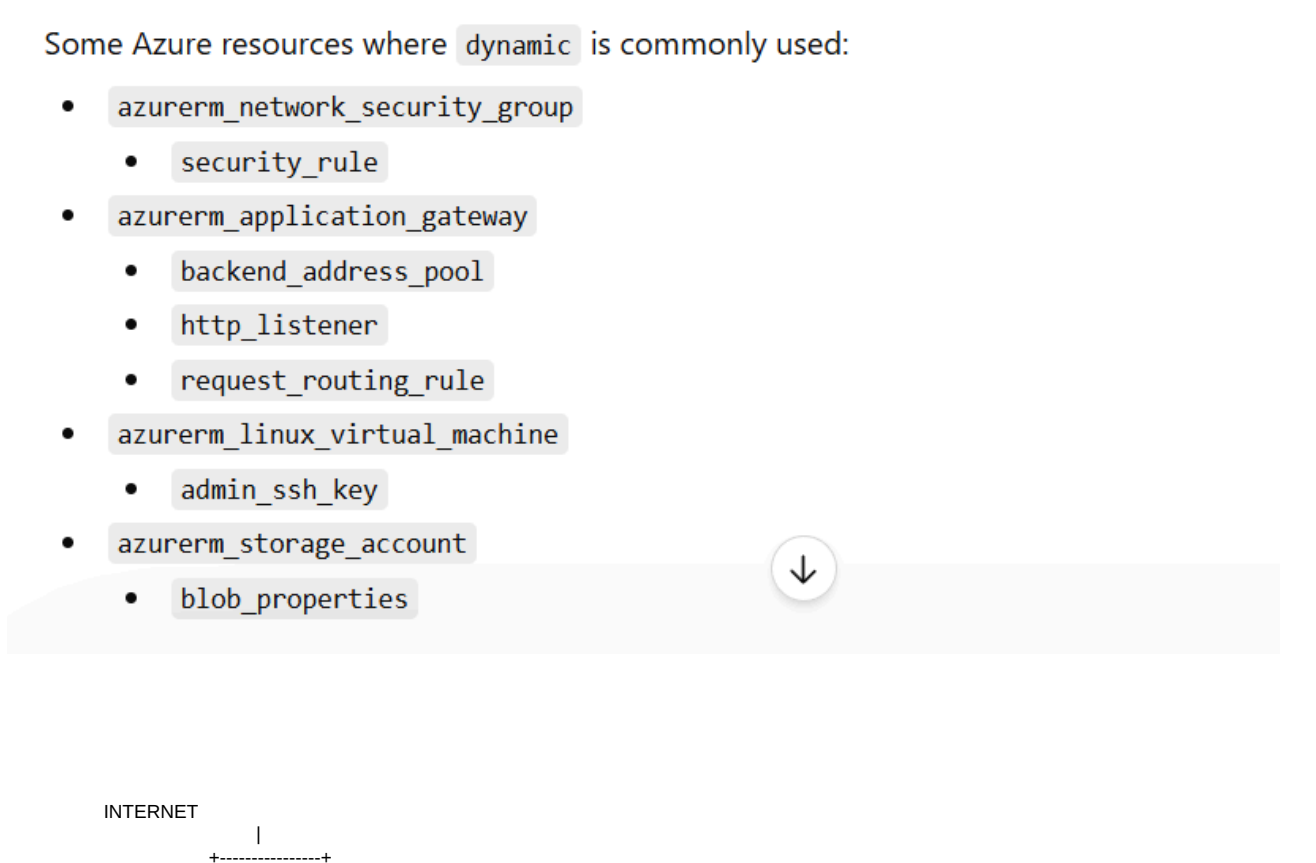
10.1.4.0
192.16.8.1
172.8.4.2
600.256.400.0

192.16.8.4

2^4 2^3 2^2

1 0 1 0 0 0 0 0

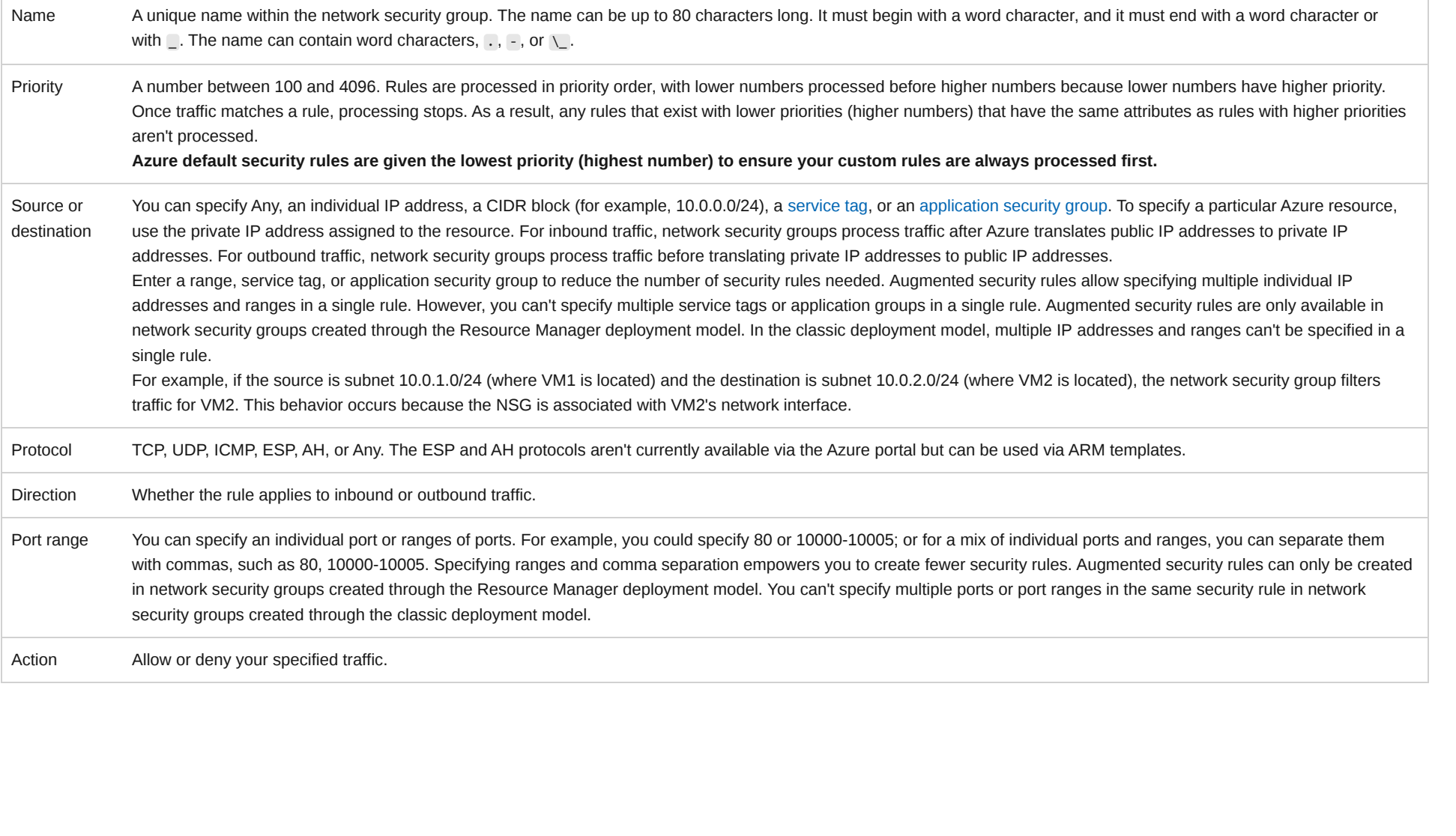
2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0 = 255



Common Azure Resources Using Dynamic Blocks

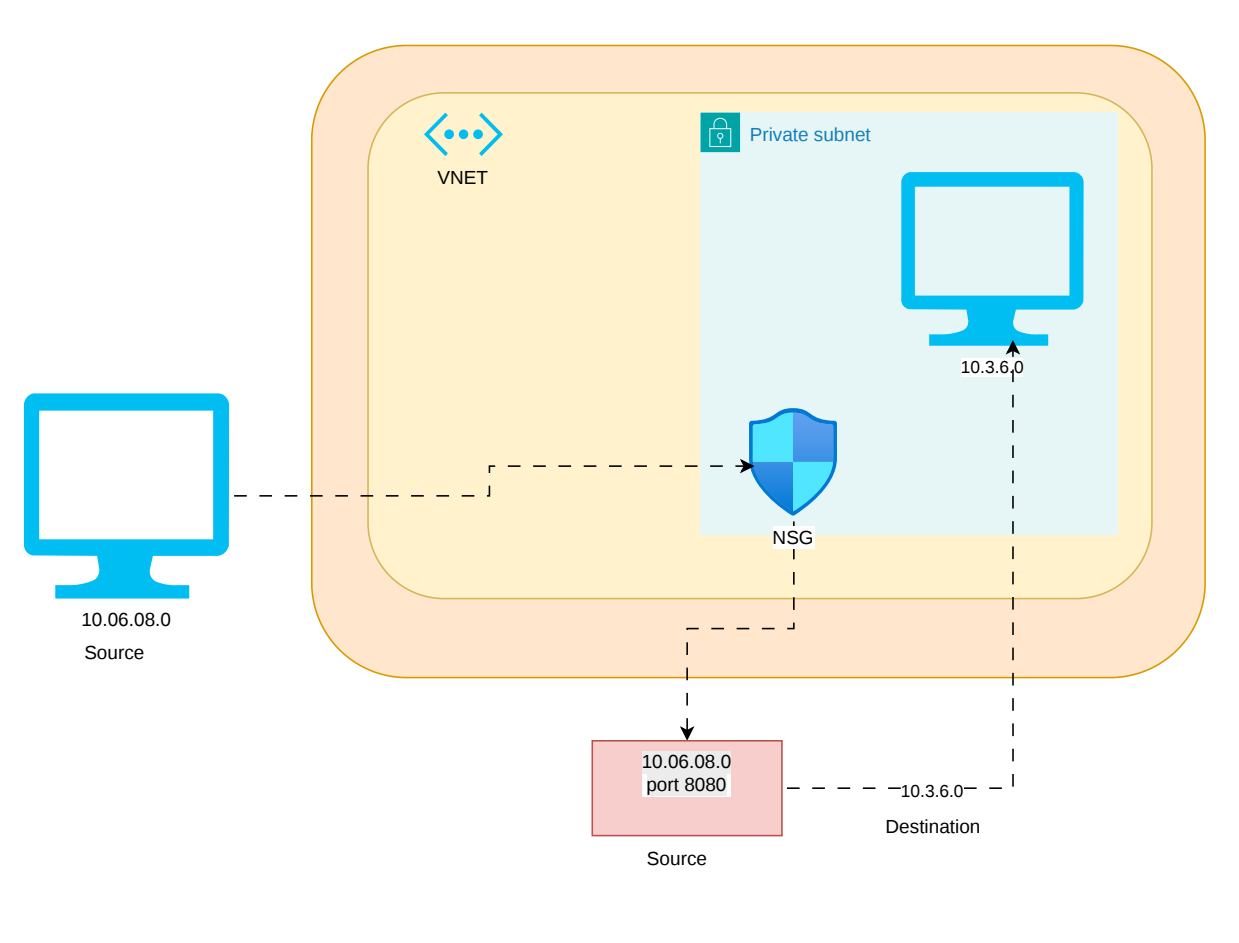
Some Azure resources where `dynamic` is commonly used:

- `azurerm_network_security_group`
  - `security_rule`
- `azurerm_application_gateway`
  - `backend_address_pool`
  - `http_listener`
  - `request_routing_rule`
- `azurerm_linux_virtual_machine`
  - `admin_ssh_key`
- `azurerm_storage_account`
  - `blob_properties`



Property	Explanation
Name	A unique name within the network security group. The name can be up to 80 characters long. It must begin with a word character, and it must end with a word character or with `_`. The name can contain word characters, `.`, `-`, or `\_`.
Priority	A number between 100 and 4096. Rules are processed in priority order, with lower numbers processed before higher numbers because lower numbers have higher priority. Once traffic matches a rule, processing stops. As a result, any rules that exist with lower priorities (higher numbers) that have the same attributes as rules with higher priorities aren't processed. **Azure default security rules are given the lowest priority (highest number) to ensure your custom rules are always processed first.**
Source or destination	You can specify Any, an individual IP address, a CIDR block (for example, 10.0.0.0/24), a service tag, or an application security group. To specify a particular Azure resource, use the private IP address assigned to the resource. For inbound traffic, network security groups process traffic after Azure translates public IP addresses to private IP addresses. For outbound traffic, network security groups process traffic before translating private IP addresses to public IP addresses. Enter a range, service tag, or application security group to reduce the number of security rules needed. Augmented security rules allow specifying multiple individual IP addresses and ranges in a single rule. However, you can't specify multiple service tags or application groups in a single rule. Augmented security rules are only available in network security groups created through the Resource Manager deployment model. In the classic deployment model, multiple IP addresses and ranges can't be specified in a single rule. For example, if the source is subnet 10.0.1.0/24 (where VM1 is located) and the destination is subnet 10.0.2.0/24 (where VM2 is located), the network security group filters traffic for VM2. This behavior occurs because the NSG is associated with VM2's network interface.
Protocol	TCP, UDP, ICMP, ESP, AH, or Any. The ESP and AH protocols aren't currently available via the Azure portal but can be used via ARM templates.
Direction	Whether the rule applies to inbound or outbound traffic.
Port range	You can specify an individual port or ranges of ports. For example, you could specify 80 or 10000-10005; or for a mix of individual ports and ranges, you can separate them with commas, such as 80, 10000-10005. Specifying ranges and comma separation empowers you to create fewer security rules. Augmented security rules can only be created in network security groups created through the Resource Manager deployment model. You can't specify multiple ports or port ranges in the same security rule in network security groups created through the classic deployment model.
Action	Allow or deny your specified traffic.

```
INTERNET
                 |
        +----------------+
        | Public Traffic |
        +----------------+
                 |
                 v
        +----------------+
        | Azure Firewall |
        |   / Gateway    |
        +----------------+
                 |
                 v
=========================================================
|                 Virtual Network (VNet)                |
|                                                       |
|  +-----------------------------------------------+   |
|  |                 Subnet-A                      |   |
|  |                                               |   |
|  |   Network Security Group (NSG)                |   |
|  |   -----------------------------------------   |   |
|  |   Rule 100: Allow TCP 80                      |   |
|  |   Rule 110: Allow TCP 443                     |   |
|  |   Rule 120: Deny All                          |   |
|  |   -----------------------------------------   |   |
|  |                                               |   |
|  |     +-----------------------+                 |   |
|  |     | Virtual Machine / App |                 |   |
|  |     | 10.0.1.4              |                 |   |
|  |     +-----------------------+                 |   |
|  +-----------------------------------------------+   |
|                                                       |
|  +-----------------------------------------------+   |
|  |                 Subnet-B                      |   |
|  |                                               |   |
|  |     Different NSG Rules                       |   |
|  +-----------------------------------------------+   |
|                                                       |
=========================================================
```

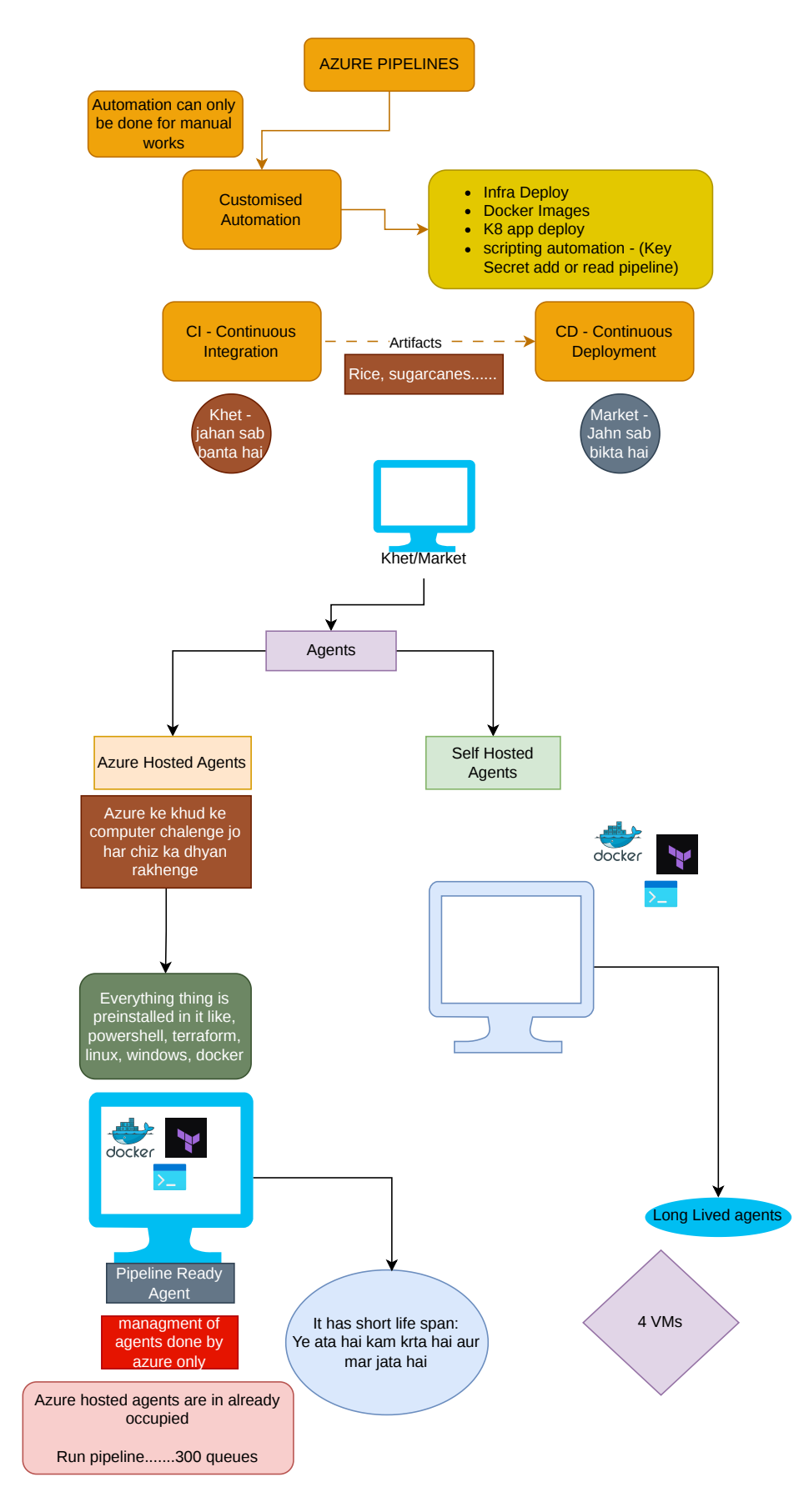

VNET
Private subnet
10.3.6.0
10.06.08.0
Source
NSG
10.06.08.0
port 8080
Source
10.3.6.0
Destination

Root Module

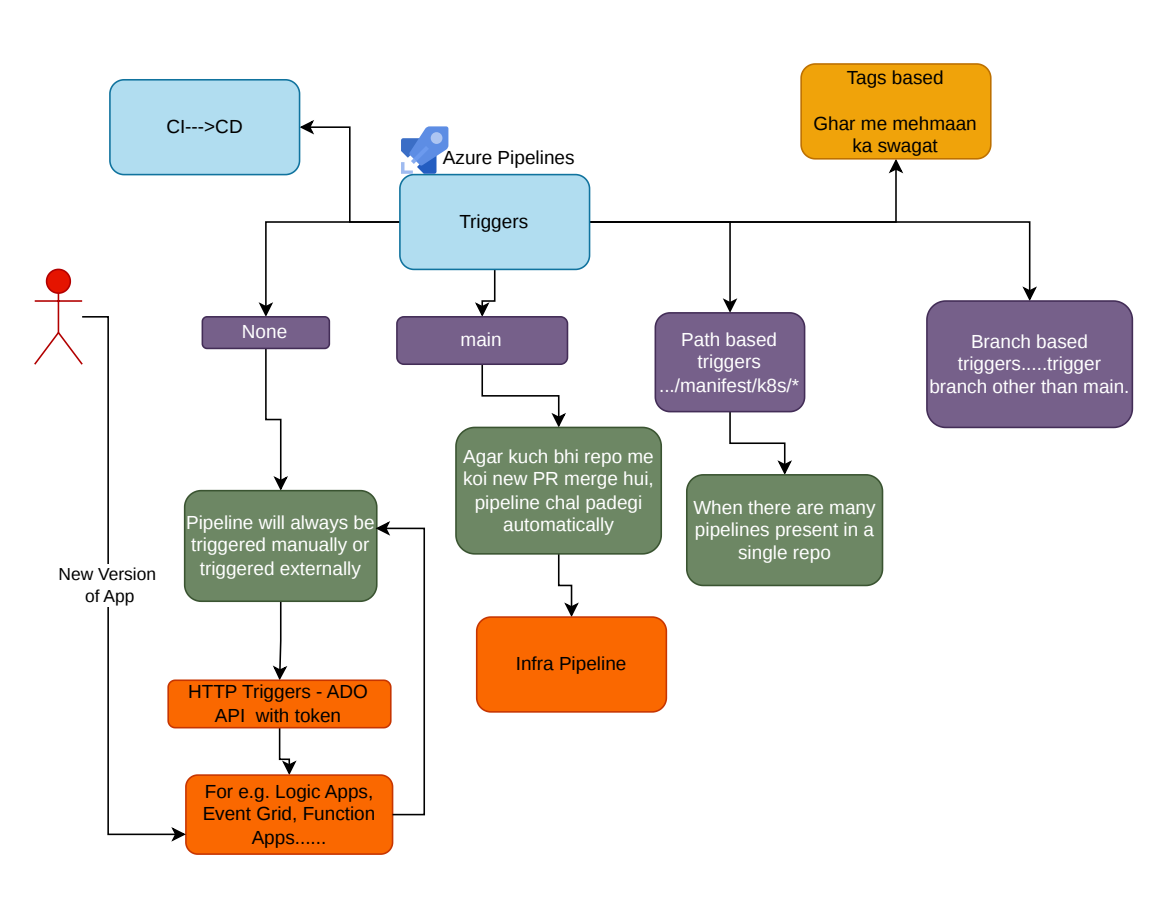
provider.tf
main.tf
var.tf
terraform.tfvars

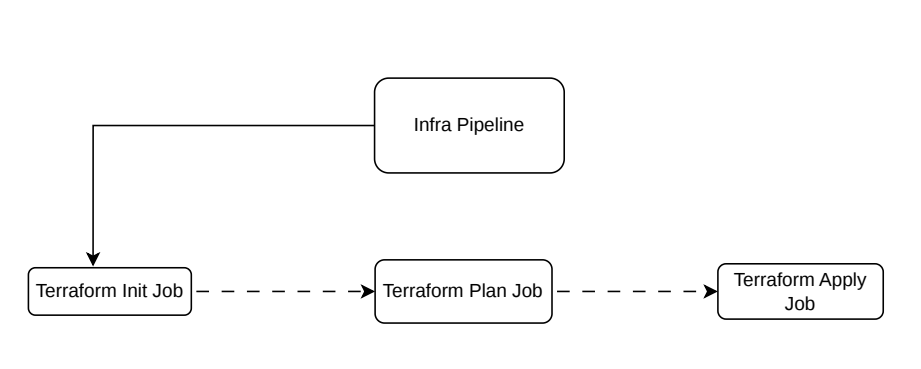
Child Module

main.tf
var.tf
data.tf


AZURE PIPELINES
Automation can only be done for manual works
Customised Automation
CI - Continuous Integration
Artifacts
Rice, sugarcanes......
CD - Continuous Deployment
Khet - jahan sab banta hai
Market - Jahn sab bikta hai
Khet/Market
Agents
Azure Hosted Agents
Azure ke khud ke computer chalenge jo har chiz ka dhyan rakhenge
Everything thing is preinstalled in it like, powershell, terraform, linux, windows, docker
docker
Pipeline Ready Agent
managment of agents done by azure only
It has short life span: Ye ata hai kam krta hai aur mar jata hai
Azure hosted agents are in already occupied
Run pipeline.......300 queues
Self Hosted Agents
docker
Long Lived agents
4 VMs

- Infra Deploy
- Docker Images
- K8 app deploy
- scripting automation - (Key Secret add or read pipeline)


CI--->CD
Azure Pipelines
Triggers
Tags based
Ghar me mehmaan ka swagat
None
main
Path based triggers ../manifest/k8s/*
Branch based triggers.....trigger branch other than main.
New Version of App
Pipeline will always be triggered manually or triggered externally
Agar kuch bhi repo me koi new PR merge hui, pipeline chal padegi automatically
When there are many pipelines present in a single repo
HTTP Triggers - ADO API with token
Infra Pipeline
For e.g. Logic Apps, Event Grid, Function Apps......


Infra Pipeline
Terraform Init Job
Terraform Plan Job
Terraform Apply Job